

Insulin Pump Therapy in Type 2 Diabetes

Andrew Tran, PharmD, BCPS

Clinical Pharmacist

Learning Objectives

- By the end of this presentation, the audience should be able to:

- ☐ Understand the role of insulin pumps in the management of Type 2 diabetes.
- ☐ Determine appropriate candidates for pump therapy.
- ☐ Interpret and utilize pump data in practice to optimize glycemic control.

Introduction to Patient

- LG is a 57 y/o male who presents to the clinic on 01/04/19 for follow-up on T2DM managed with insulin pump. He reports testing 2x/day and does not eat consistent meals.
- PMH:
 - T2DM
 - Peripheral neuropathy
 - Hypertension
 - Dyslipidemia
 - Hypothyroidism
 - TIA
 - PVD
 - BPH
- FH: History of heart disease on both sides.
- SH: Current every day smoker. Smokes 1 PPD X 14 years.
- Insurance: Medicare

Patient Case

Cont'd

- Vitals: 10/27/2018
 - BP: 114/68 mmHg
 - Ht: 72 in.
 - Wt: 120 kg (265 lb)
- A1c: 6.4 (10/19/2018)
- ASCVD 10-year risk: 20.8%
- Lipid Panel (10/27/2018)
 - TC: 148
 - HDL: 33
 - LDL: 80
 - TG: 175
- Urine microalbumin: 89.0 mg/g (11/23/2018)
- SCr: 1.32 (10/29/2018)
- EGFR: 56 min/hr (10/29/2018)
- CrCl: 105 mL/min (10/29/2018)

Patient Case

Medication Name	Directions	Indication
Novolog U-100 Vial	Administer insulin via pump. 1 vial lasts 9 days.	T2DM
Metformin 1000 mg tablet	Take 1 tablet by mouth twice daily	
Gabapentin 600 mg tablet	Take 1 tablet by mouth twice daily	Peripheral neuropathy
Metoprolol tartrate 50 mg tablet	Take 1 tablet by mouth twice daily	Hypertension
Amlodipine 5 mg tablet	Take 1 tablet by mouth once daily for blood pressure	
Pravastatin 40 mg tablet	Take 1 tablet by mouth every night for cholesterol	Dyslipidemia
Fenofibrate 135 mg capsule	Take 1 tablet by mouth daily for elevated triglycerides	
Aspirin 81 mg tablet	Take 1 tablet by mouth daily	Stroke prophylaxis
Tamsulosin 0.4 mg capsule	Take 1 capsule by mouth nightly	BPH
Synthroid 175 mcg tablet	Take 1 tablet by mouth 30 minutes before breakfast	Hypothyroidism

Insulin Pump Overview

- Background
- Methodology
- Clinical
Guidelines
- Eligibility
- Pros and Cons

Introduction

- **n** Diabetes is characterized by hyperglycemia due to decreased insulin secretion, decreased insulin sensitivity, or BOTH.
 - Type 1 Diabetes (T1DM) – Insulin Deficiency
 - Type 2 Diabetes (T2DM) – Insulin Resistance

Introduction

In T1DM, multiple daily insulin injections (MDIs) are used to achieve glycemic control.

- Patients with T2DM typically progress to intensive insulin therapy after lack of control with oral therapies



1. American Diabetes Association. Standards of Medical Care in Diabetes—2020. Diabetes Care. 2020;43(suppl 1):S1-S212.

2. Image: <https://www.mcguiffmedical.com/content/images/thumbs/0010662_05cc-insulin-syringe-29g-x-12-safety-bd-safetyglide-100box.jpeg>

3. Image: <<https://getcertified.mvflextouch.com/Content/images/flexpens.png>>

Basal vs. Bolus Insulin

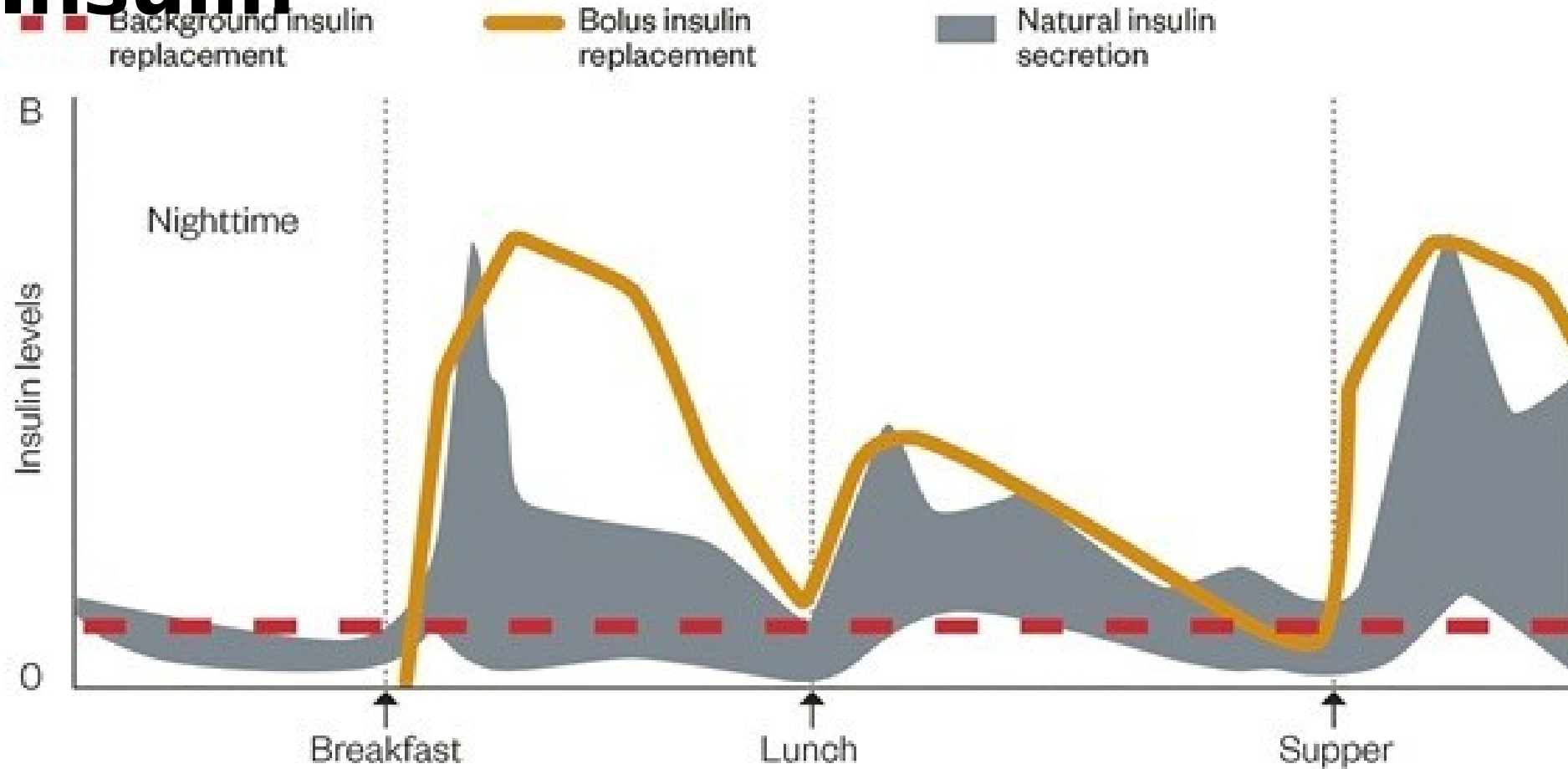
Basal Insulin	Bolus Insulin
• Provides consistent insulin level throughout the day • Decreases fasting blood glucose • Intermediate or long-acting insulin • Approximately 50% of total daily dose	• Administered to decrease glucose after meals (post-prandial) • Rapid or short-acting insulin • Approximately 50% of total daily dose

Insulin

Dosing

- Daily basal insulin therapy
 - 0.1-0.2 units/kg/day (ABW)
 - OR
 - 10 units/daily
- Multi-dose insulin therapy (4 injections daily)
 - 0.3-0.5 units/kg/day
 - 50% basal insulin
 - 50% bolus insulin divided with meals

Insulin Therapy Mimics Natural Insulin

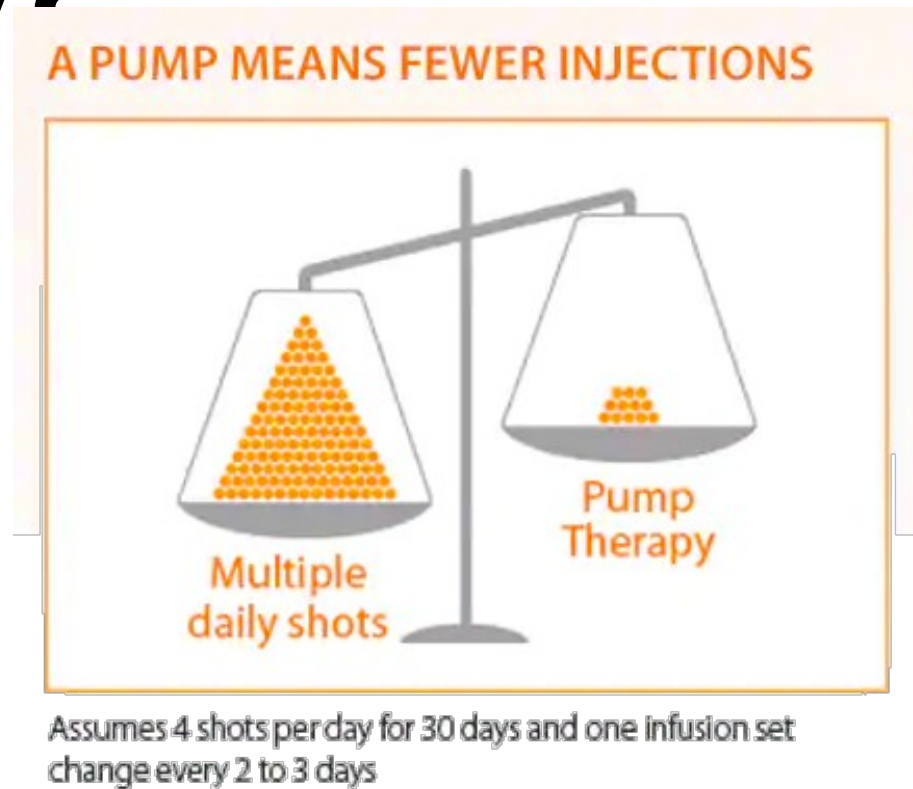


Introduction to Insulin

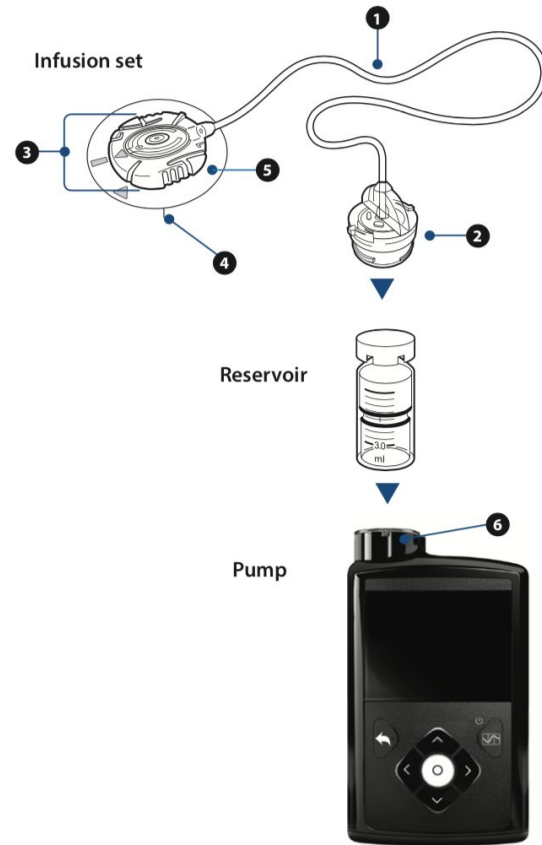
Pumps

- Continuous subcutaneous insulin infusion (CSII) or insulin pump therapy, is a method of insulin delivery for T1DM or T2DM patients.
- It is estimated that approximately 30-40% of T1DM patients utilize insulin pumps in the U.S.
- There are fewer people with T2DM that use insulin pumps. However, this is on the rise.

Advantages to Pump therapy?



Appearanc



1. <https://www.medtronicdiabetes.com/treatments/insulin-pump-therapy-type-2>
2. <https://www.medtronicdiabetes.com/get-started>

How They Work

- Mimic physiologic insulin release by providing continuous insulin administration throughout a 24-hour period.
- Pumps deliver continuous amounts of rapid-acting insulin (e.g. lispro, aspart, glulisine) that replicate basal insulin secretion.
- While bolus delivery only occurs when activated by the user to cover carbohydrate consumption.



Previous Meta-

Reference	Findings
Weissberg-Benchell et al, <i>Diabetes Care</i> . 2003;26(4):1079-1087	Compared with MDI, CSII therapy was associated with significant improvements in glycemic control based on HbA_{1c} and mean blood glucose decreases
Jeitler et al, <i>Diabetologia</i> . 2008;51(6):941-951	HbA_{1c} reduction greater and insulin requirements lower with CSII than with MDI in adults and adolescents with T1DM; hypoglycemia risk comparable among adult patients (data unavailable for adolescent subjects); no conclusive CSII benefits for patients with T2DM
Fatourechi et al, <i>J Clin Endocrinol Metab</i> . 2009;94(3):729-740	In patients with T1DM, HbA_{1c} was mildly decreased with CSII vs. MDI; CSII effect on hypoglycemia unclear; similar CSII and MDI outcomes among patients with T2DM
Pickup and Sutton, <i>Diabet Med</i> . 2008;25(7):765-774	HbA_{1c} was lower for CSII than for MDI, with greatest improvement in patients with highest initial HbA_{1c} values on MDI; severe hypoglycemia risk was decreased with CSII vs. MDI; greatest reduction in patients with diabetes of longest duration and/or highest baseline rates of severe hypoglycemia
Monami et al, <i>Exp Clin Endocrinol Diabetes</i> . 2009;117(5):220-222	HbA_{1c} was significantly lower with CSII vs. MDI; HbA_{1c} reduction was only evident for studies with mean patient age >10 years; severe hypoglycemia occurred at comparable rates with CSII and MDI therapy

Opt2mise Trial (2014)

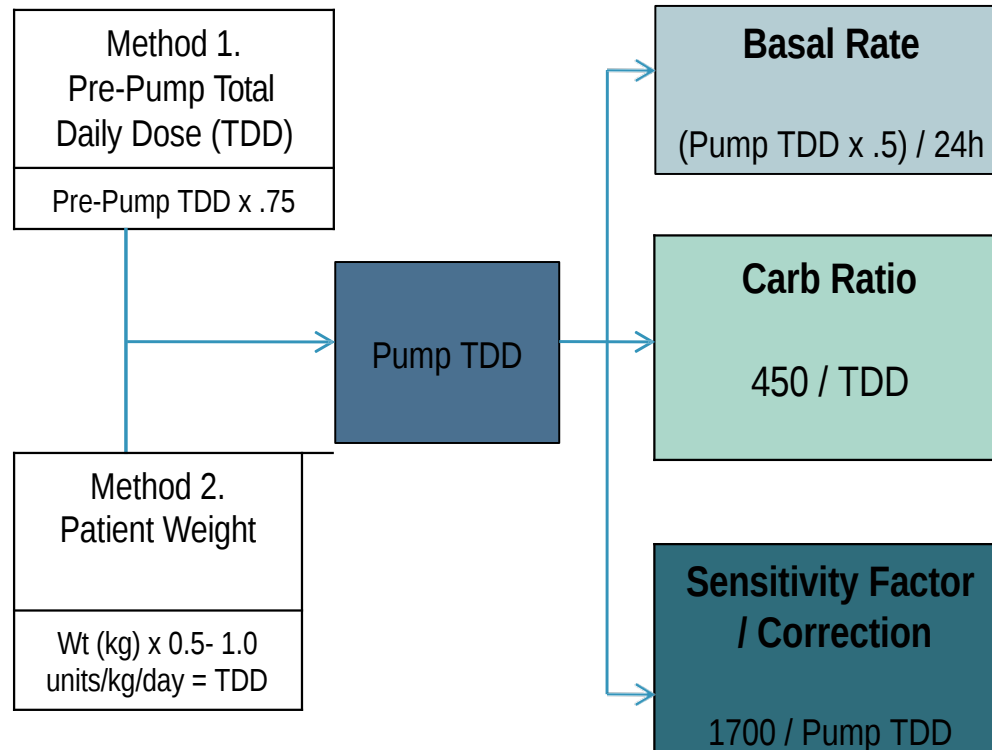
Overview	Primary Outcome	Inclusion Criteria
Multi-center, randomized, controlled, parallel-group with single-arm crossover Objective: Compare efficacy and safety of pump therapy vs MDIs in T2DM patients who have failed to respond to a basal-bolus regimen. 331 patients randomized Duration: 6 months	The primary end point is the between-group difference in mean change in HbA1c from baseline to 6 months.	Insulin-taking patients with type 2 diabetes on MDI (>3 injections/day) $8.0\% \leq A1C \leq 12.0\%$ Insulin dose: 0.7-1.8 units/kg/day Mean SMBG ≥ 2.5 times/day
Primary Outcome		
- After 6 months, HbA1c levels decreased by 1.1% in the pump therapy group and by 0.4% in the multiple daily injections group ($P < .001$). - Compared with participants in the injections group, participants in the pump therapy group were more likely to achieve HbA1c levels below 8% ($P < .001$)		

Candidates for Insulin Pump Therapy

AACE Proposed Clinical Characteristics of Suitable Insulin Pump Candidates		Notes
Class 1	Class 2	Class 3
Patients with T1DM who do not reach glycemic goals despite adherence to maximum MDI, non-CSII program, especially if they have: -Very labile DM (erratic and wide glycemic excursions, including recurrent DKA) -Frequent severe hypoglycemia and/or hypoglycemic unawareness -Significant "dawn phenomenon," extreme insulin sensitivity -Special populations (e.g., preconception, pregnancy, children and adolescents with eating disorders, competitive athletes)	Patients with T1DM who are on a maximized basal-bolus MDI insulin regimen, regardless of their level of glycemic control and who, after investigation and careful consideration, feel that CSII would be helpful or more suitable for lifestyle reasons	Selected patients with insulin-requiring T2DM who satisfy any or all of the following: • C-peptide positive, but with suboptimal control on a maximal program of basal/bolus injections • Substantial "dawn phenomenon" • Erratic lifestyle (e.g., frequent long-distance travel, shift-work, unpredictable schedules leading to difficulty maintaining timing of meals) • Severe insulin resistance, candidate for U500 insulin by CSII • Selected patients with other DM types (e.g., post-

Calculations for Insulin Pump Settings

Clinical Guidelines



-Start with 1 basal rate, adjust according to glucose trends over 2-3 days
 -Adjust to maintain stability in fasting state (between meals & during sleep)
 -Add additional basals according to diurnal variation (dawn phenomenon)

-Adjust based on low-fat meals with known carbohydrate content
 -Acceptable 2-h post-prandial rise is ~60mg/dL above pre-prandial BG
 -Adjust carb ratio in 10%-20% increments based on post-prandial BG

ALTERNATE METHODS

-Carb Ratio: (6x Wt in kg / TDD) or (2.8 x Wt in lbs / TDD)
 -Fixed Meal Bolus = (TDD x .5) / 3 equal meals (not carb counting)

Clinical Considerations on Pump TDD

- Average values from Method 1 & 2
- Hypoglycemic patients □ start at lower value
- Hyperglycemic, elevated A1C, or pregnant □ start at higher value

TDD: total daily dose BG: blood glucose

-Sensitivity Factor is correct if BG is within 30 mg/dL of target range within 2 hours after correction

Eligibility and Coverage

- Medicare
 - Patient must:
 - Test BGs 4x/day for at least 60 days
 - Self-injecting insulin at least 3x/day for 6 months
 - CCS Certificate of Medical Necessity (CMN)
 - Concurrent Labs:
 - C-peptide within 110% of lower range
 - Fasting BG < 225 mg/dL

Pros and Cons

Advantages	Disadvantages
Less injections	Price
Precise and more predictable insulin delivery	Steep learning curve (testing, use of pump settings)
Tighter blood glucose control	Has to be worn all the time (skin irritation)
Reduced hypoglycemic episodes	Increased risk of DKA if pump malfunctions

Back to the Patient

History of

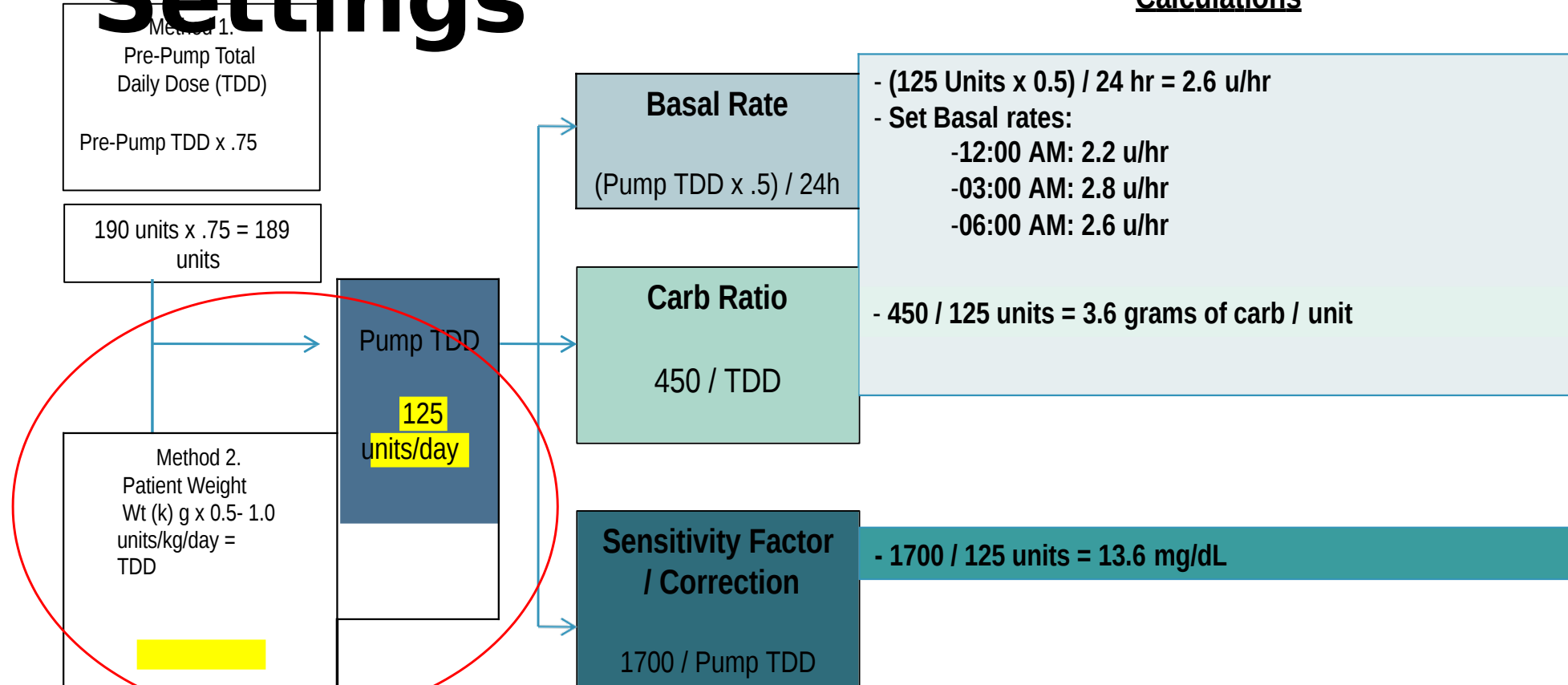
10-24-2017 - Pharmacy Diabetes visit

- Diabetes Medications:
 - **Levemir 100 U/mL Flextouch** - Inject 54 units in the morning and 57 units in the evening
 - **Novolog Flexpen 100 U/mL** - Administer 20 units AM, 32 units at Noon, and 27 units with supper
 - Metformin 1000 mg - Take 1 tablet by mouth twice daily
- POC A1C: 11.1%; BGs: 200s-300s
- Insulin TDD = 190 units
- Recently started seeing dietician for carb counting.
- Joint visit with PCP to recommend insulin pump.
- Patient agrees to begin process.
- Started demo with saline.

01-12-2018 - Pharmacy Diabetes visit

- Met with Pharmacist and Medtronic representative to discuss pump features and set-up.
- Discontinue old insulin regimen.
- Started pump at insulin TDD 125 units.

Calculations for Insulin Pump Settings



Clinical Considerations on Pump TDD

- Average values from Method 1 & 2 = 125 units/day
- Hypoglycemic patients ☐ start at lower value
- Hyperglycemic, elevated A1C, or pregnant ☐ start at higher value

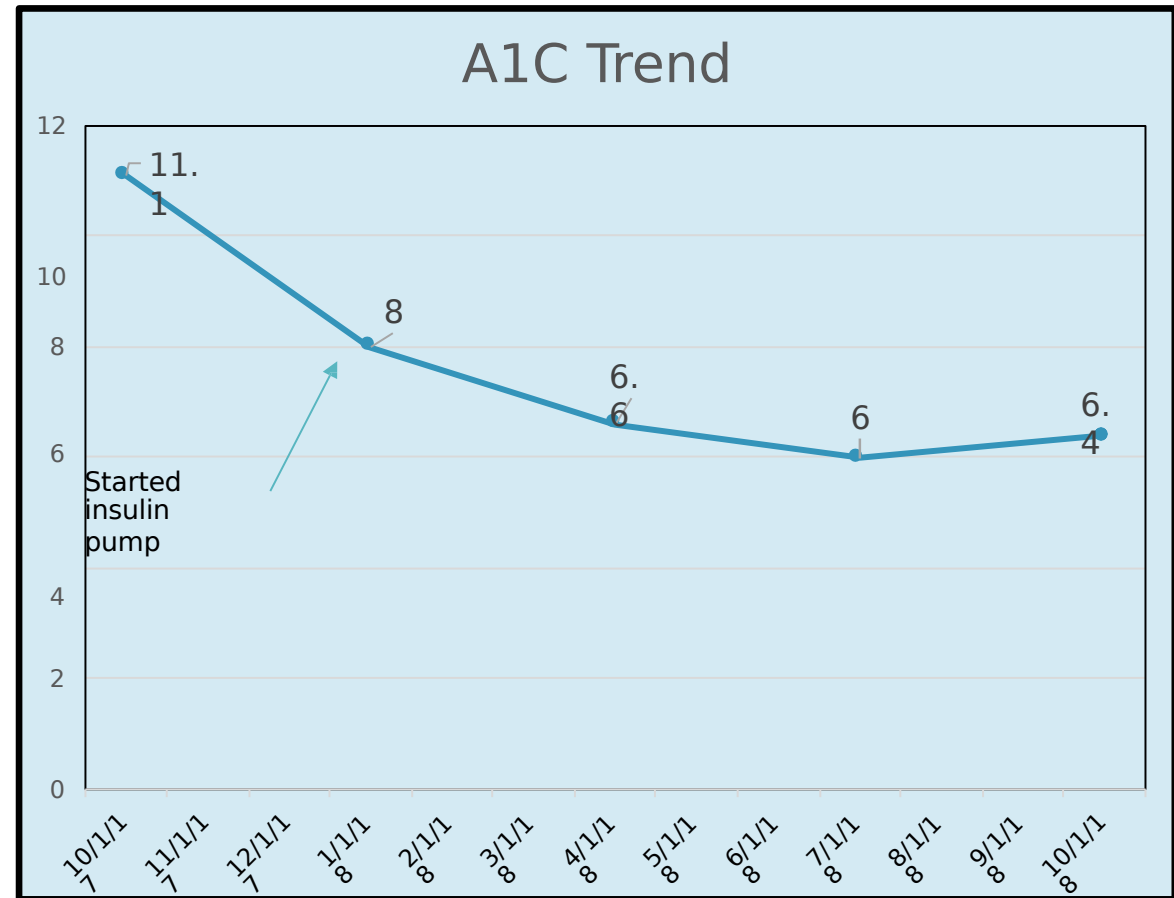
AACE/ACE Consensus Statement, Endocr Pract. 2014;20(No.5)

TDD: total daily dose BG: blood glucose

Patient Case

Cont'd

Date	A1C (%)
10/24/17	11.1
1/26/18	8
4/24/18	6.6
7/26/18	6
10/19/18	6.4



Back to 01/04/2019

Visit

- LG is a 57 y/o male who presents to the clinic on 01/04/19 for follow-up on T2DM managed with insulin pump. He reports testing 2x/day and does not eat consistent meals.
- PMH:
 - T2DM
 - Peripheral neuropathy
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- Insurance: Medicare

LG's Current Insulin

R

Bolus	
Carbohydrate Ratio (g/U)	10.0
Insulin Sensitivity (mg/dL per U)	15.0
Blood Glucose Target (mg/dL)	100

Basal Rate	
Time	Units/Hour
00:00	1.80
03:00	2.30
06:00	1.90
12:00	1.90
19:00	1.80
24-hour Basal total: 45.90 Units	

Patient Case

Cont'd

SMBG (g/dL)														
	12/21	12/22	12/23	12/24	12/25	12/26	12/27	12/28	12/29	12/30	12/31	01/01	01/02	01/03
Breakfast		120		168		130	113	81			110	149	87	
Lunch			86 60		93		70	68	137					55
Dinner	146	96	66	77		59	81			154	162	172	151	115

Total Daily Insulin Used														
	12/21	12/22	12/23	12/24	12/25	12/26	12/27	12/28	12/29	12/30	12/31	01/01	01/02	01/03
Insulin (Units)	49.6	46.3	45.9	51.1	46.0	48.6	47.4	45.9	49.0	50.1	52.1	55.4	50.00	47.6

Statistics	12-21 - 01-03
Avg BG (mg/dL)	108
Avg Daily Carbs (g)	12
Avg Daily Insulin (units)	48.93
Avg Daily Bolus (units)	3.04

Assessment

- Type 2 Diabetes
 - Well-controlled based on last A1C of 6.4%, which is at goal of < 7.0% per ADA.
 - SMBG values are mostly at goal. However, there are several lows after lunch.

Pla

•nType 2 Diabetes

- Decrease 12:00-7:00 basal rate by 10% to 1.7 units/hour.
- Consider adding ACEi/ARB therapy due to proteinuria and switch to high intensity statin due to ASCVD 10 year-risk > 7.5%, if tolerated.
- Carb/counting and SMBG education discussed.
- Follow-up in 2 weeks.

Basal Rate	
Time	Units/Hour
00:00	1.80
03:00	2.30
06:00	1.90
12:00	1.90
19:00	1.80
24-hour Basal total: 45.90 Units	



Basal Rate	
Time	Units/Hour
00:00	1.80
03:00	2.30
06:00	1.90
12:00	1.70
19:00	1.80
24-hour Basal total: 44.60 Units	

Take-Home Points

- Insulin pumps may be a viable option for glucose control for T2DM patients, who must have intensive insulin therapy.
- When considering appropriate candidates for insulin pump, patient specific factors must be considered.
- Proper training and education are essential for success.

Questions

• Feel free to contact me with any questions:

Andrew Tran, PharmD,
BCPS Clinical Pharmacist
Baylor Scott & White
Health Andrew.Tran@utexas.edu

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